

Toluwalase Ifebajo Design Document

This is a Sokoban puzzle game created with .NET MAUI.

To accomplish each level, simply push boxes onto the target squares. I chose this over the music player because the game logic caught interest, and the level editor seemed like a fun addition.

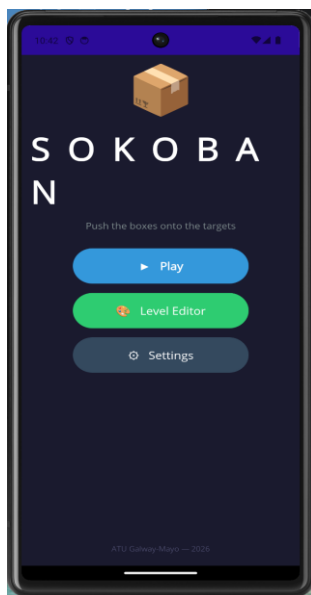
The app operates on both Mac and Android and contains five pages:

The main menu includes Level Select, Game, Level Editor, and Settings.

App Structure

The software navigates between pages using Shell.

- **Main Menu:** Select Play, Editor, or Setting



Level Select: Opens the selected level. **Game Page** ← Saves progress after each level is completed.

- **Level Editor** → Test levels on Game Page.

Levels, progress, and settings are all recorded as JSON on the device. Levels are downloaded from GitHub during the initial run.

To keep the code tidy, I separated it into **Models, Services, ViewModels, and Views**. The **GameEngine** manages game movement independently of the UI.



Decisions

Why I used C# for the grid: The grid size varies across levels, thus I constructed it dynamically in C# rather than XAML.

Why I utilised a Stack for Undo: Using a stack speeds up the undo process. I save the grid before each move and then restore it when the undo button is pushed.

Why I utilised GitHub for levels: Levels are downloaded from GitHub on the first run and saved locally. I have included eight levels as a backup in case the download fails.

Why I utilised Kenney sprites: Kenney sprites enhance the game's visual appeal and are cross-platform compatible.

Why did I add arrow buttons: Swipe controls can be unstable, therefore buttons ensure that the game is always playable.

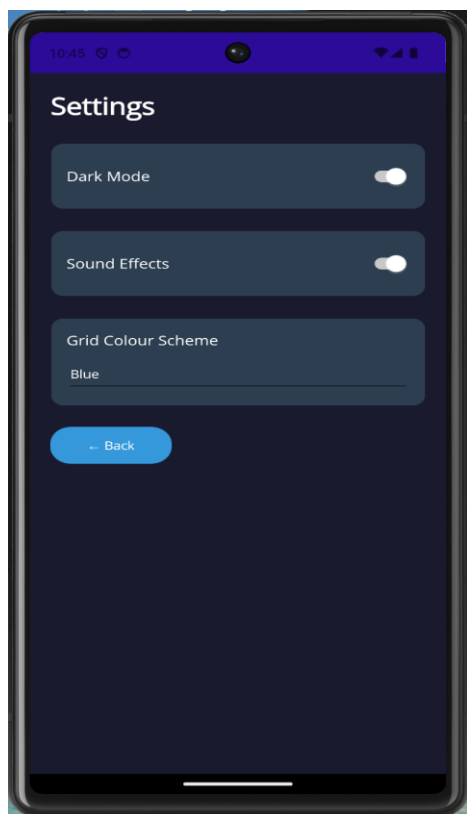
What Works: All eight built-in levels load and are playable.

- Mac users can move the player using arrow buttons, while Android users can swipe and pan.
- Undo and reset both functions properly.

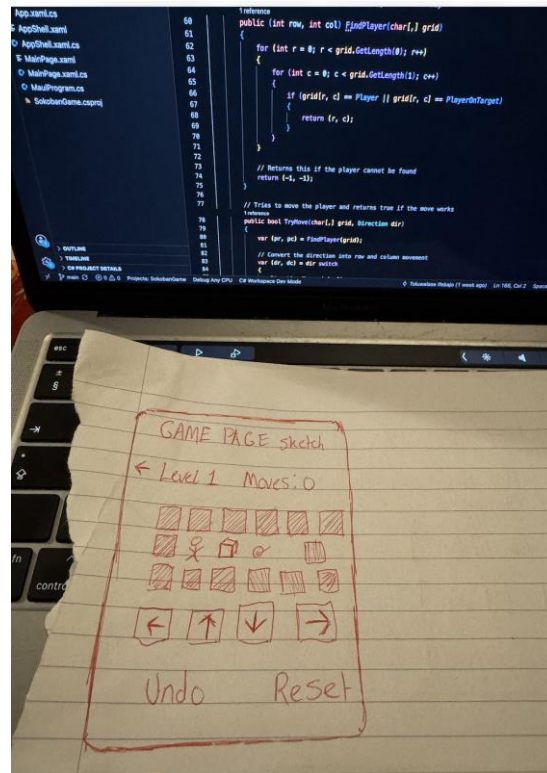
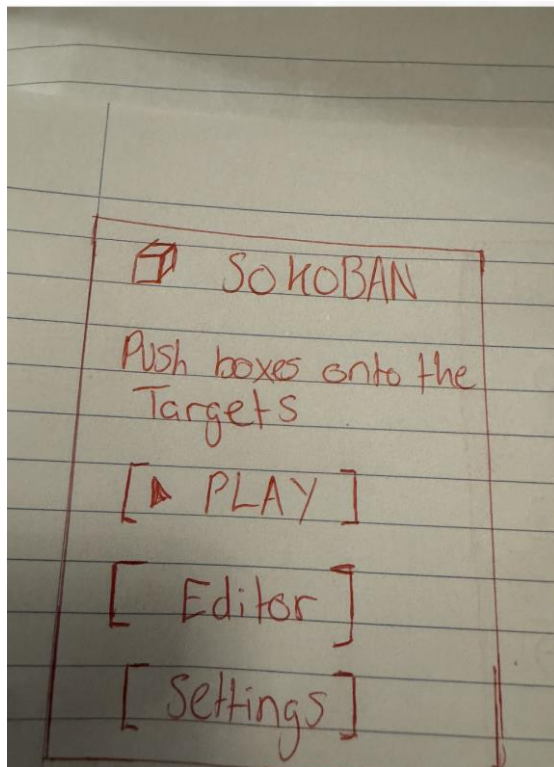
- Win detection ensures all boxes are on targets and saves the highest score
- Level choice displays completed levels with ticks and greatest move count
- Level editor: You can paint a grid and modify its size
- Test before saving your work start the level in game mode
- Custom levels appear in the level choose with a star
- Settings: dark mode shifts promptly and saves
- Sound toggle saves and loads properly
- The app works on both Mac and Android

What Doesn't Work

- I didnt get to add animations for movement or win screen due to time constraints
- Sound effects are set up but theres no actual audio files added yet so the toggle doesnt play anything



Sketches



[settings]

SELECT LEVEL

[Lv1] [Lv2] [Lv3]

[Lv4] [Lv5] [Lv6]

[Lv7] [Lv8] [MYLv]